

ON A MONOTONICITY PROPERTY OF PILLAI-NANDA TRACE TEST

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Summary: By using the theory of vanishing Jacobians, a method is provided to determine a critical point of the power functions of multivariate normal analysis likelihood ratio test. A multivariate noncentral distribution is proved to be monotonic increasing, showing that Pillai-Nanda trace test is monotonic nondecreasing.

1. INTRODUCTION

Several statistical optimization problems involve determining global optimum values of a given function on the boundaries of given regions. Calculus methods sometimes do fail to give such end point extrema. Thus, e.g., the minimum of

$$(1) \quad \phi(\rho) = \rho/(1-\rho), \quad 0 \leq \rho < 1,$$

occurs at $\rho = 0$. However, the derivative of (1) equated to zero does not yield $\rho = 0$.

To determine end point extrema, when calculus methods fail, statisticians

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use several other methods. Amongst them, fixed point theory methods, mathematical programming theory methods, functional and structural relationships (dependence) optimization methods, and establishing multivariate normal probability inequality methods are prominent. Thus, e.g., Srivastava and Khatri (1979, pp. 299-304) use Anderson's (1955) probability inequality to determine monotonicity of some likelihood ratio tests. The importance of monotonicity lies in the fact that it ensures unbiasedness of a test. The test region, although convex, is not necessarily convex symmetric about the origin. However, several of the known multivariate probability inequalities established in the literature for symmetric convex regions about the origin do hold good for nonsymmetric or for nonconvex symmetric regions. One such known inequality is Sheppard's result. For convenience we record it here.

For the bivariate normal distribution with zero means, unit variances, and correlation coefficient $\rho > 0$, we have from Sheppard's result that

$$(2) \quad \phi(\rho) = P\{x, y > 0\} = \left[\frac{\pi}{2} + \tan^{-1}(\rho/(1-\rho^2)^{1/2}) \right] / 2\pi.$$

Obviously, $\phi(\rho)$ is monotonic increasing and hence has a minimum at $\rho = 0$, and thus we observe that

$$(3) \quad P\{x, y > 0 | \rho > 0\} > P\{x, y > 0 | \rho = 0\}.$$

Again from (2), notice that

$$(4) \quad P\{x, y \in R | \rho > 0\} > P\{x, y \in R | \rho = 0\},$$

where R denotes the region of the first and third quadrants. This region although symmetric, about the origin, is nonconvex.

In section 3 it is proved using (2) that

$$(5) \quad P\{x, y \in R | \rho > 0\} > P\{x, y \in R | \rho = 0\},$$

where R is any convex region in the first quadrant. The result (5) shows

that Theorem 2 of Shafer and Olkin (1983) remains true over other regions than the convex symmetric about origin regions assumed by them. Theorem 2 of Shafer and Olkin (1983) states as follows. Let x, y be two k -component column vectors such that $(x'y')'$ has a $2k$ -variate normal distribution with mean vector zero, and $2k \times 2k$ correlation matrix

$$(6) \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} = \begin{bmatrix} R & J_k' J_k - R \\ J_k J_k' - R & R \end{bmatrix},$$

where J_k is a column vector of k unities, then

$$(7) \quad P(x, y) = P\{|x_1| < c, \dots, |x_k| < c, |y_1| < c, \dots, |y_k| < c\}$$

is minimized when all correlations in R equal $\frac{1}{2}$, and c is a known constant. For a generalization of (6), (7) see Kabe (1985).

Thus to prove monotonicity properties of the likelihood ratio tests, we must develop probability inequalities over arbitrary convex regions, i.e. the rejection regions determined by the tests.

In this paper we use vanishing Jacobians theory to extend Anderson's (1955) inequality to certain rejection regions, and then use this inequality to establish a monotonicity property of a noncentral multivariate beta distribution, which in turn establishes the monotonicity property of Pillai-Nanda trace test, which Srivastava and Khatri (1979, section 10.3.1, p. 304) remark as only partially known.

Some known results are stated in the next section and the main results of the paper are given in section 3.

2. SOME KNOWN RESULTS

Suppose we wish to optimize $\phi(y)$ subject to $g(y) = a$, where y has n

components and a has $m \leq n - 1$ components. Then an optimal y_0 satisfies $g(y_0) = a$ and the following system of $(n-m)$ vanishing Jacobian equations

$$(8) \quad \begin{vmatrix} \frac{\partial \phi}{\partial y_k} & \frac{\partial \phi}{\partial y_1} & \frac{\partial \phi}{\partial y_2} & \cdots & \frac{\partial \phi}{\partial y_m} \\ \frac{\partial g_1}{\partial y_k} & \frac{\partial g_1}{\partial y_1} & \frac{\partial g_1}{\partial y_2} & \cdots & \frac{\partial g_1}{\partial y_m} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ \frac{\partial g_m}{\partial y_k} & \frac{\partial g_m}{\partial y_1} & \frac{\partial g_m}{\partial y_2} & \cdots & \frac{\partial g_m}{\partial y_m} \end{vmatrix} = 0,$$

$k = m + 1, \dots, n$ and the cofactor of $\partial \phi / \partial y_k$ is assumed to be nonzero. The Hessian matrix

$$(9) \quad H = \begin{bmatrix} \frac{\partial^2 \phi(y)}{\partial y \partial y'} & \frac{\partial g'(y)}{\partial y} \\ \frac{\partial g(y)}{\partial y'} & 0 \end{bmatrix}_{y=y_0},$$

determines the nature of the optimum. Let H_1 be the determinant of H , H_i be the determinant obtained from H by deleting its first $(i-1)$ rows and columns, $i = 1, \dots, m$. Then $\phi(y_0)$ is a minimum if $H_1, \dots, H_m$ all have the sign of $(-1)^m$, $\phi(y_0)$ is a maximum if the signs of $H_1, \dots, H_m$ alternate. The result (8) is mentioned by Trench (1978, pp. 458-459).

Let x be a p -component column vector, S a symmetric $p \times p$ positive definite matrix, then $x'Sx \leq c$ is a convex set. By writing $S = A'A$, $Ax_1 = y_1$, $Ax_2 = y_2$, we note that

$$(10) \quad 2\left(\frac{y_1 + y_2}{2}\right)' \left(\frac{y_1 + y_2}{2}\right) \leq y_1'y_1 + y_2'y_2,$$

which satisfies the definition of convexity. It follows that

$$(11) \quad u \leq x'Sx \leq u + du,$$

is a convex symmetric about origin set. Further, $XGX' = M$, X $p \times n$, G $n \times n$ symmetric positive definite, is a convex set. $|M| < c$ and $\text{tr}M < c$ are also convex sets.

3. MAIN RESULTS

We first proceed to generalize (3). Let a $p \times n$ matrix X have a pn -variate normal distribution with mean vector zero, and a $pn \times pn$ positive definite correlation matrix equal to the covariance matrix $\Sigma \otimes I$. Then

$$(12) \quad P\{X > 0 | \Sigma\} > P\{X > 0 | \Sigma = I\},$$

where $X > 0$ indicates that all elements of X are positive. Here we optimize

$$(13) \quad P\{X > 0 | \Sigma\} = \phi(\alpha_{12}, \dots, \alpha_{1p}, \dots, \alpha_{p-1, p}, \bar{\alpha}_{12}, \dots, \bar{\alpha}_{p-1, p}),$$

subject to the $p(p-1)/2$ conditions.

$$(14) \quad \phi_{ij} = \alpha_{ij} + \bar{\alpha}_{ij} = 1, \quad i, j = 1, \dots, p; \quad i \neq j, \quad i < j,$$

where ρ_{ij} of Σ is $\rho_{ij} = \alpha_{ij} - \bar{\alpha}_{ij}$.

In all we have $[p(p-1)/2 + 1]$ functions of $p(p-1)$ variates $\alpha_{12}, \dots, \bar{\alpha}_{p-1, p}$. If $X = (x_1, \dots, x_n)$, then (12) reduces to

$$(15) \quad \prod_{i=1}^n P\{x_i > 0 | \Sigma\} > \prod_{i=1}^n P\{x_i > 0 | \Sigma = I\}.$$

Thus it is sufficient to prove that

$$(16) \quad P\{x > 0 | \Sigma\} > P\{x > 0 | \Sigma = I\},$$

where x is any arbitrary column vector of X .

First consider a proof of (3). Now $\rho > 0$ can be expressed as $\rho = \alpha - \bar{\alpha}$, $\alpha + \bar{\alpha} = 1$, and hence (3) can be verified by using Lagrange multiplier theory. However, (5) is difficult to verify by Lagrange multiplier theory, and so we write (5) as

$$(17) \quad \text{Optimize } \phi_1(\rho) = \phi_1(\alpha - \bar{\alpha}) = P\{x, y \in R | \rho > 0\},$$

subject to

$$(18) \quad \phi_2 = \alpha + \bar{\alpha} = 1.$$

Now we have two functions of two variates each, and hence from (8)

$$(19) \quad \begin{vmatrix} \frac{\partial \phi_1}{\partial \alpha} & \frac{\partial \phi_1}{\partial \bar{\alpha}} \\ 1 & 1 \end{vmatrix} = \frac{\partial \phi_1}{\partial \alpha} - \frac{\partial \phi_1}{\partial \bar{\alpha}} = 0,$$

and $\alpha = \bar{\alpha} = \frac{1}{2}$ or $\rho = 0$ is a critical point. This point gives a minimum because from Sheppard's result $\phi(\rho)$, $\rho > 0$, is monotonic nondecreasing over the entire first octant.

We now proceed to prove (16). We know that (16) is true for $p = 2$. We assume it true for $(p-1)$ and then for a general p we prove that

$$(20) \quad \phi(\alpha_{1p}, \dots, \bar{\alpha}_{p-1p}) = P\{x_{pp} > 0 | x_{p-1} > 0, \Sigma\},$$

subject to the $(p-1)$ conditions

$$(21) \quad \phi_i = \alpha_{ip} + \bar{\alpha}_{ip} = 1, \quad i=1, \dots, p-1,$$

is optimized at $\alpha_{1p} = \bar{\alpha}_{1p} = \dots = \bar{\alpha}_{p-1p} = \frac{1}{2}$, where x_{pp} is the p -th component of x and x_{p-1} is the vector of the first $(p-1)$ components of x .

Now we have p functions of $2(p-1)$ variates each and hence from (8) their Jacobian system of $(p-1)$ Jacobians vanishes. This system is

$$(22) \quad \begin{vmatrix} \frac{\partial \phi}{\partial \bar{\alpha}_{ip}} & \frac{\partial \phi}{\partial \alpha_{1p}} & \frac{\partial \phi}{\partial \alpha_{2p}} & \dots & \frac{\partial \phi}{\partial \alpha_{p-1,p}} \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \cdot & \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \cdot & \dots & \cdot \\ 1 & 0 & 0 & \dots & 0 \\ \cdot & \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \cdot & \dots & \cdot \\ 0 & 0 & 0 & \dots & 1 \end{vmatrix} = 0$$

$$= \frac{\partial \phi}{\partial \bar{\alpha}_{ip}} - \frac{\partial \phi}{\partial \alpha_{ip}}, \quad i = 1, \dots, p-1.$$

Thus $\alpha_{ip} = \bar{\alpha}_{ip}$ or $\rho_{ij} = 0$ is an optimal point. Since (16) is minimized for $p=2$, we conclude that (16) is minimized for any general p .

In view of the results so far established we state, without proof, a generalization of Anderson's (1955) inequality.

Let $p \times n$ X with $E(X) = \mu$ and covariance matrix $\Sigma \otimes I$ have a pn -variate normal distribution, where all the correlations in Σ are positive. However, we assume that Σ is the correlation matrix itself. Let $K = \text{diag}(k_1, \dots, k_p)$, $0 \leq k_i \leq 1$, $i = 1, \dots, p$.

Then it holds that

$$(23) \min_{K, \Sigma} \int_{XX' = G} \exp\{-\frac{1}{2} \text{tr} \Sigma^{-1}(X - K\mu)(X - K\mu)'\} dX,$$

occurs at $K=0$, $\Sigma=I$. Note that the integral (22) represents the density of G without the normalizing constants.

We now state a monotonicity result for Hotelling's T^2 -distribution. First consider the χ^2 distribution for testing $\mu = 0$ against $\mu \neq 0$, where now x , with $E(x) = \mu$ and $E(x-\mu)(x-\mu)' = \Sigma$ is p -variate normal. From (23) we have that

$$(24) \int_c^\infty du \int_{x' \Sigma^{-1} x = u} \exp\{-\frac{1}{2}(x-\mu)' \Sigma^{-1}(x-\mu)\} dx \\ > \int_c^\infty du \int_{x' \Sigma^{-1} x = u} \exp\{-\frac{1}{2}x' \Sigma^{-1} x\} dx,$$

i.e. noncentral χ^2 density monotonically increases in the positive parameter $\mu' \Sigma^{-1} \mu$. Similarly for Hotelling's T^2 -test we know that

$$(25) P\{T^2 > c | \mu' \Sigma^{-1} \mu\} > P\{T^2 > c | \mu' \Sigma^{-1} \mu = 0\}.$$

A generalization of (25) to the multivariate beta distribution establishes the monotonicity property of Pillai-Nanda trace test. We proceed with this generalization.

It follows from (23) that

$$(26) \min_K \int_{YY' = G_2, XX' = G_1} \exp\{-\frac{1}{2} \text{tr} \Sigma^{-1}[(X-K\mu)(X-K\mu)' + YY']\} dX dY,$$

occurs when $K=0$. Here X is $p \times n_1$ and Y is $p \times n_2$. We now write (26) as

$$(27) \min_K \int_{(XX' + YY')^{-\frac{1}{2}} XX' (XX' + YY')^{-\frac{1}{2}} = M} \exp\{-\frac{1}{2} \text{tr} \Sigma^{-1}[(X-K\mu)(X-K\mu)' + YY']\} dX dY,$$

occurs when $K=0$. However, (27) says that for $K=1$,

$$(28) \quad f(M, \Sigma^{-1} \mu \mu') \geq f(M),$$

where $f(M, \Sigma^{-1} \mu \mu')$ is the noncentral multivariate beta density of M , and $f(M)$ is the central density of M . Note that (28) is an exact generalization of (25). It follows that

$$(29) \quad \int_{\text{tr} M > c} f(M, \Sigma^{-1} \mu \mu') dM > \int_{\text{tr} M > c} f(M) dM,$$

i.e. the left-hand side of (29) is a monotonic increasing function of each root of $\Sigma^{-1} \mu \mu'$. The result (29) establishes the monotonicity property of Pillai-Nanda trace test, to test $\mu = 0$, against $\mu \neq 0$, when Σ is unknown. We reject H_0 if $\text{tr} M > c$. Further, since all the noncentral multivariate normal sampling densities are of the type (23) or (27) they are all monotonic increasing functions of each root of the noncentrality parameter matrix.

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